

Customer Shopping Behavior Analysis

A comprehensive analysis of **3,900 transactions** across product categories — uncovering spending patterns, customer segments, and strategic opportunities.

DATA ANALYSIS

PYTHON · SQL · POWER BI



Dataset at a Glance

3,900

Transactions

Total purchase records analyzed

18

Features

Customer, product & behavior fields

37

Missing Values

In Review Rating — imputed by category median

50

Locations

Geographic diversity across the dataset

Key features include demographics (Age, Gender, Location, Subscription), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

01

Load & Explore

Imported dataset via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

02

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case for consistency.

03

Feature Engineering

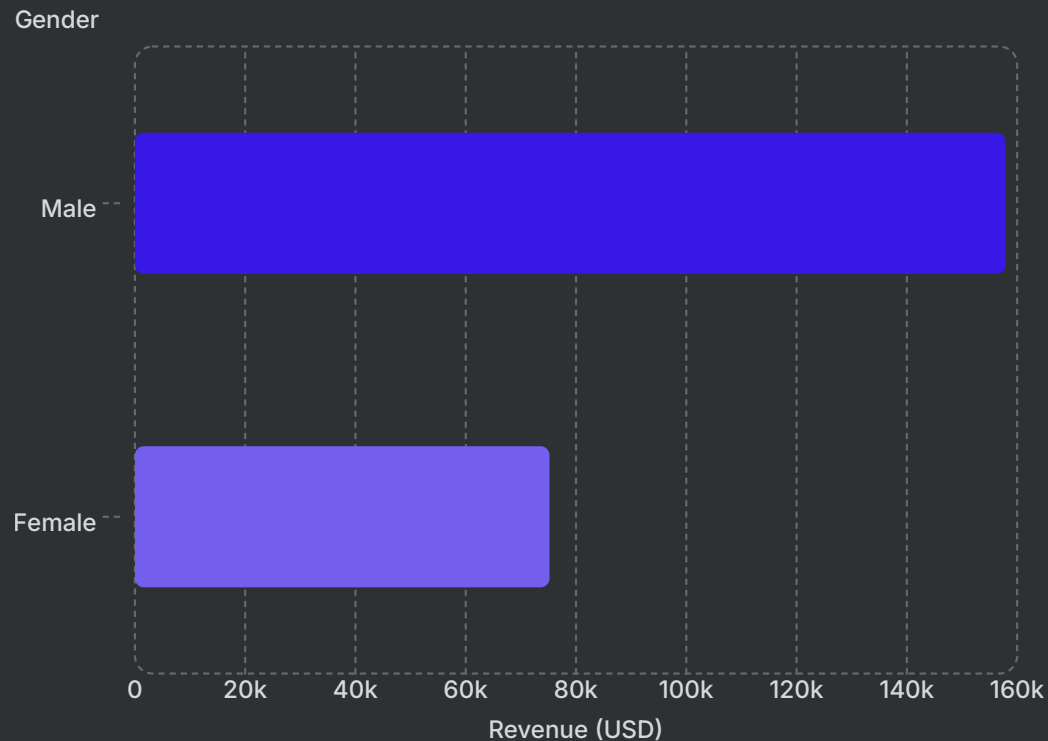
Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for structured SQL analysis.

Revenue & Spending Insights



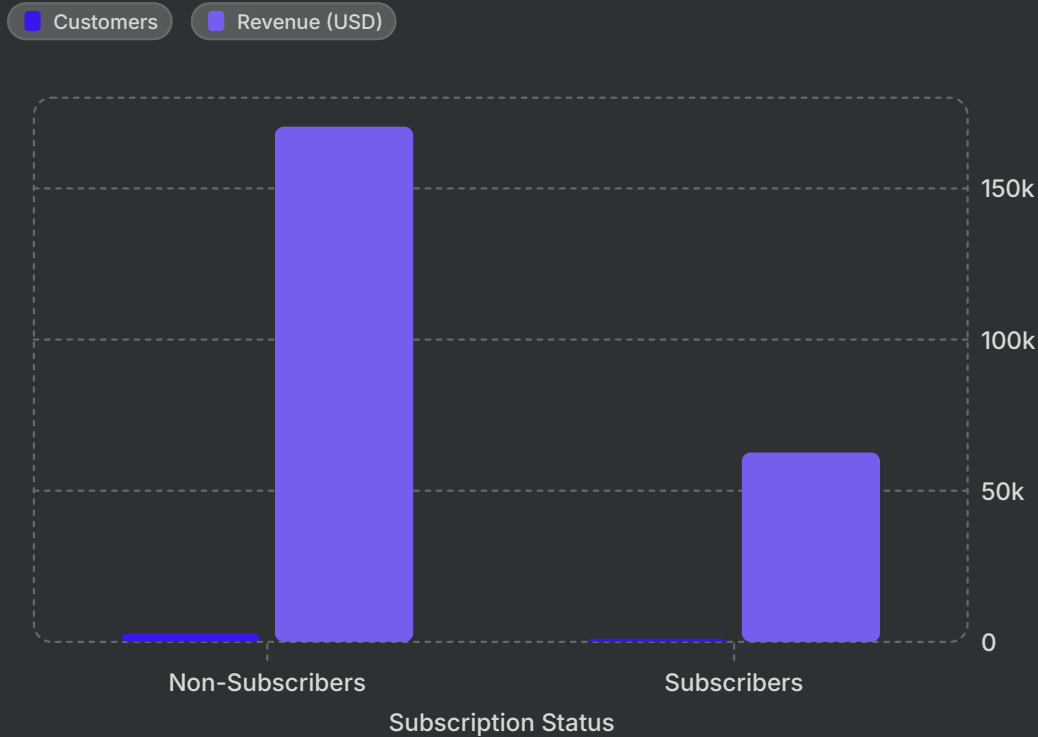
Gender Revenue Split

Male customers generated **\$157,890** in revenue — more than double female customers at **\$75,191**.

Shipping Behavior

Express shipping customers spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**), suggesting higher-intent buyers opt for faster delivery.

Subscription & Discount Analysis



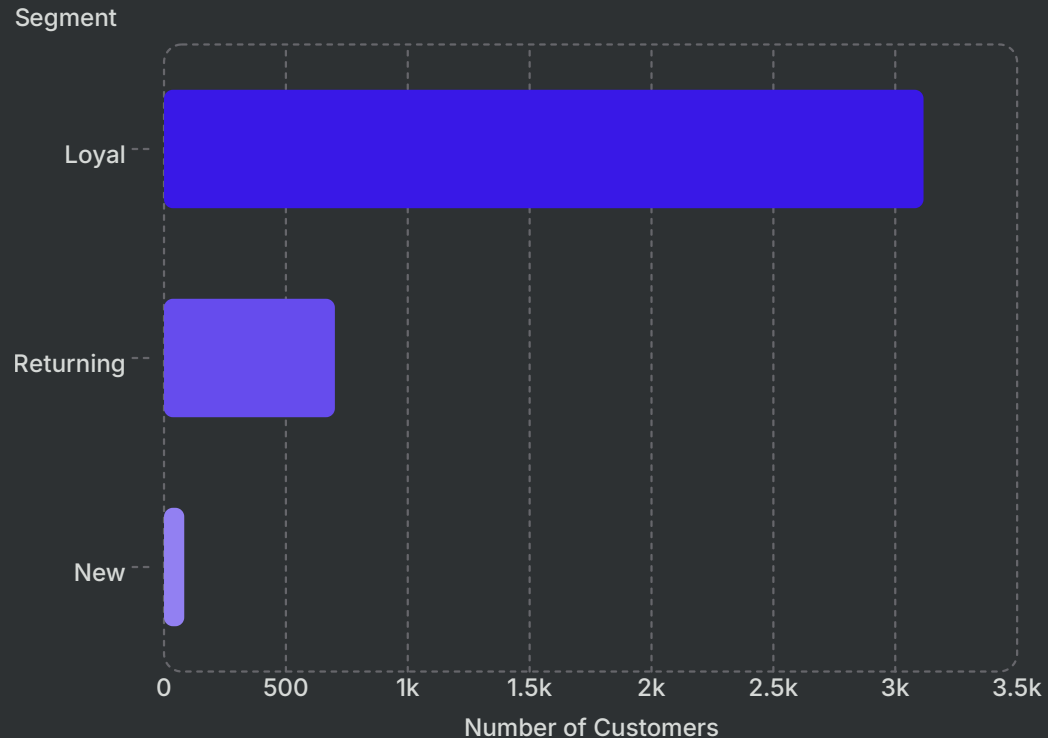
Subscribers vs. Non-Subscribers

Non-subscribers dominate at **73%** of customers and **\$170K** revenue. Average spend is nearly identical (~\$59.50), indicating strong upsell potential for subscription conversion.

Top Discount-Dependent Products

- Hat — 50.00% discounted purchases
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

Customer Segmentation



Segment Breakdown

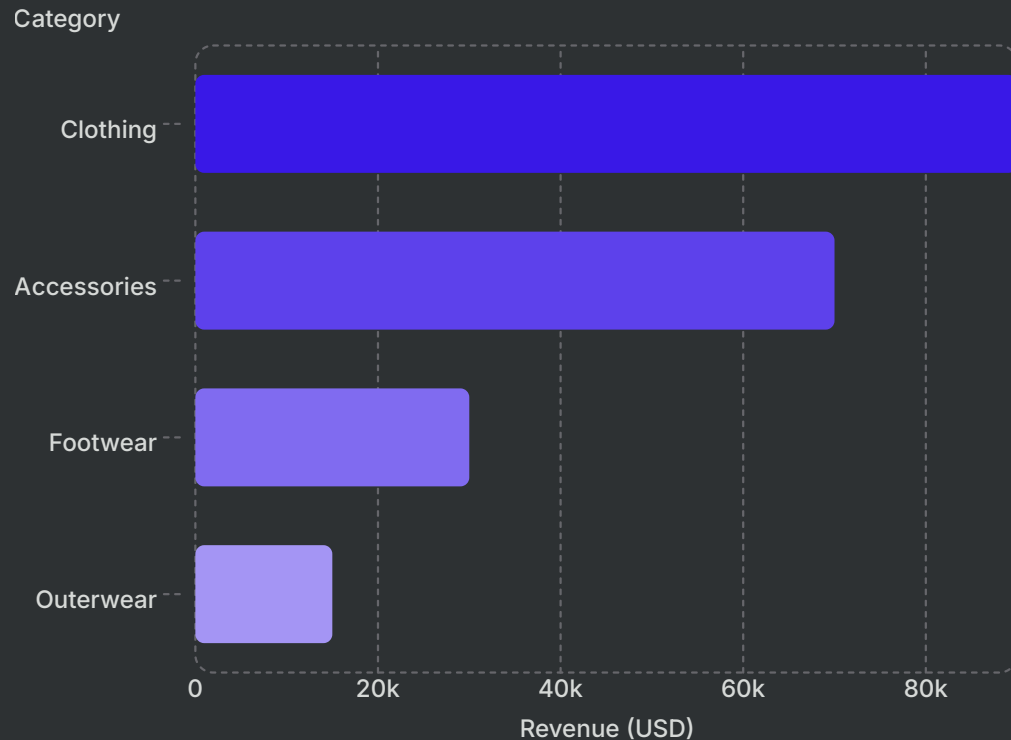
Loyal customers represent **80%** of the base (3,116 customers), followed by Returning (701) and New (83).

Repeat Buyers & Subscriptions

Of customers with **5+ purchases**, 958 are subscribers vs. 2,518 non-subscribers — a clear opportunity to convert high-frequency buyers into subscribers.

- ❑ Loyal segment is the largest — loyalty programs could further increase retention and lifetime value.

Product & Category Performance



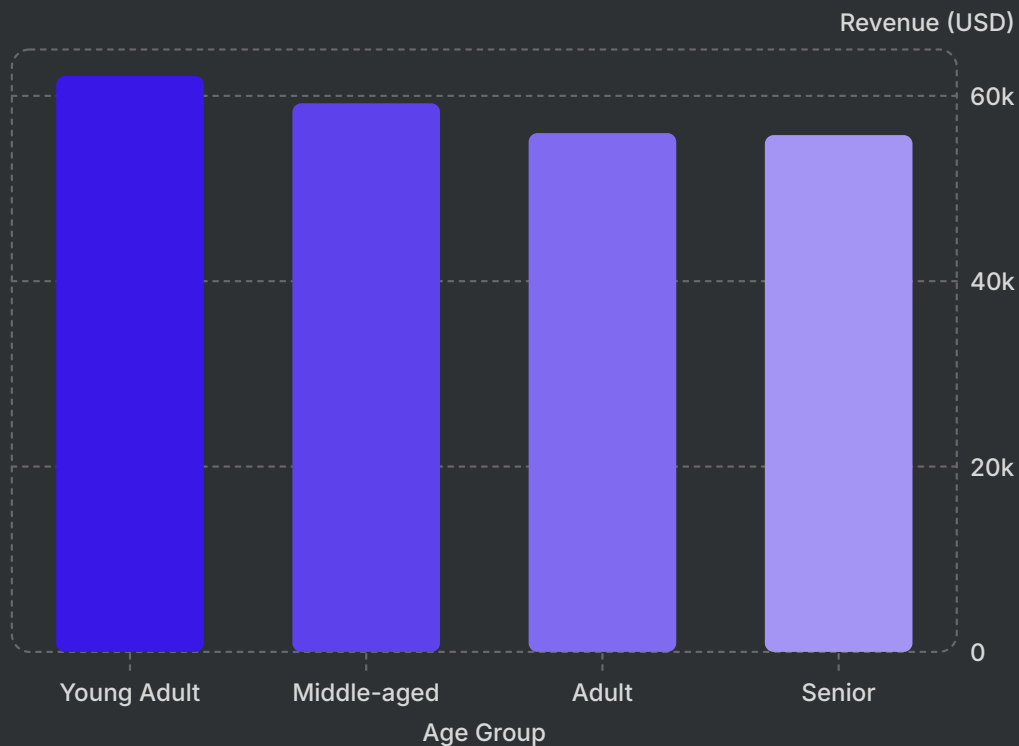
Revenue by Category

Clothing leads with ~\$90K revenue and ~1,800 sales. Accessories follows at ~\$70K. Footwear and Outerwear represent smaller but meaningful segments.

Top-Rated Products

1. Gloves — 3.86 avg rating
2. Sandals — 3.84
3. Boots — 3.82
4. Hat — 3.80
5. Skirt — 3.78

Revenue by Age Group



Age Group Revenue Contribution

Young Adults generate the highest revenue at **\$62,143**, followed closely by Middle-aged (\$59,197). Revenue is relatively evenly distributed across all age groups, with a slight decline toward Senior customers.

Key Takeaway

All four age segments contribute meaningfully to revenue — marketing should be tailored per group rather than concentrated on a single demographic.

Power BI Dashboard

An interactive **Power BI dashboard** was built to visualize all key findings with dynamic filtering by Subscription Status, Gender, Category, and Shipping Type.

3.9K Customers

Total customer base tracked

\$59.76 Avg Order

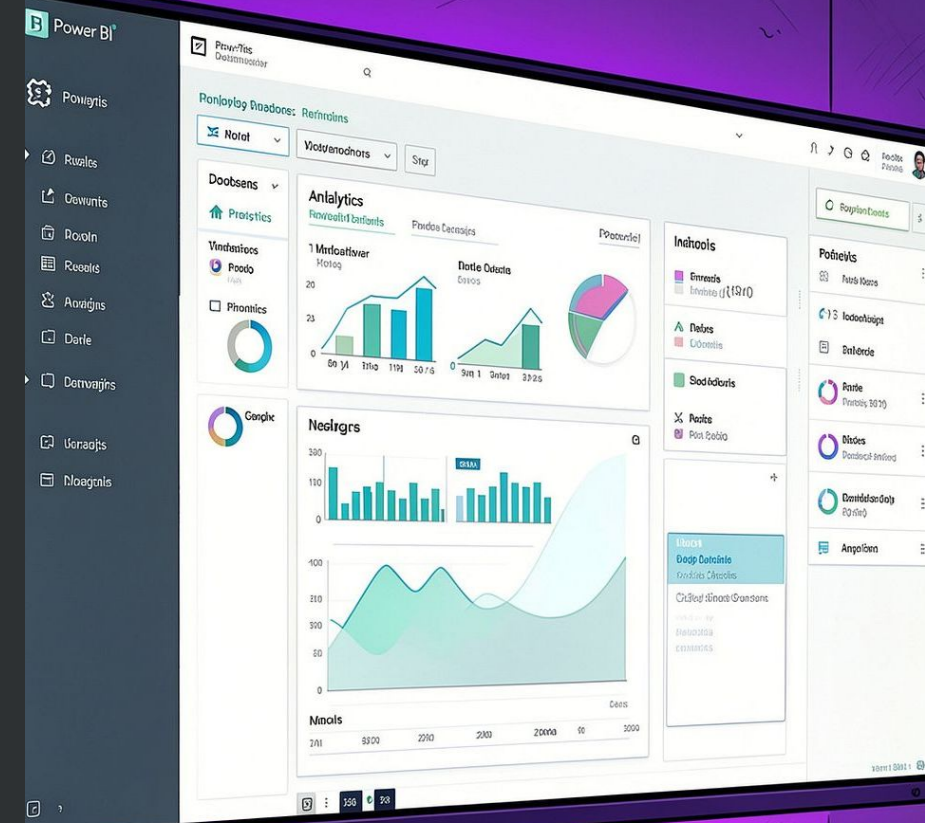
Average purchase amount

3.75 Avg Rating

Overall product review score

27% Subscribers

Subscription penetration rate



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base — especially high-frequency repeat buyers.

Loyalty Programs

Reward Returning customers to accelerate movement into the Loyal segment (currently 80% of base).

Review Discount Policy

Balance sales lift against margin erosion — Hats, Sneakers, and Coats show highest discount dependency (~49–50%).

Targeted Marketing

Focus campaigns on Young Adults (top revenue group) and Express shipping users (higher average spend).

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) and best-sellers (Blouse, Jacket, Jewelry) in marketing campaigns.